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CIRCLE

NIMCET PYQ

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By Amit Katiyar (MCA-JNU)

1. The equation of the circle passing through (2, 3) and $(-1, 1)$ whose centre lying on the line $x - 3y - 11 = 0$ is **NIMCET 2007**
(a) $x^2 + y^2 - 7x + 5y - 14 = 0$
(b) $x^2 - y^2 + 7x - 5y + 12 = 0$
(c) $x^2 + y^2 + 7x + 5y - 14 = 0$
(d) $x^2 + y^2 + 7x + 5y + 14 = 0$

2. If the lines $2x + 3y + 1 = 0$ and $3x - y - 4 = 0$ lie along diameters of a circle of circumference 10π, the equation of the circle is: **NIMCET 2007**
(a) $x^2 + y^2 + 2x + 2y - 23 = 0$
(b) $x^2 + y^2 - 2x - 2y - 23 = 0$
(c) $x^2 + y^2 - 2x + 2y - 23 = 0$
(d) $x^2 + y^2 + 2x - 2y - 23 = 0$

3. The equation of the circle passing through (2, 3) and (– 1, 1) with its centre lying on the line is $x - 3y - 11 = 0$ is (Circle) **NIMCET 2007**
(a) $x^2 + y^2 - 7x + 5y - 14 = 0$
(b) $x^2 - y^2 + 7x - 5y + 12 = 0$
(c) $x^2 + y^2 + 7x + 5y - 14 = 0$
(d) $x^2 + y^2 + 7x + 5y + 14 = 0$

4. If the lines 2x + 3y +1 = 0 and 3x – y – 4 = 0 lie along diameters of a circle of cirumference 10π ,then the equation of the circle is: (Circle) **NIMCET 2007**
(a) $x^2 + y^2 + 2x + 2y - 23 = 0$
(b) $x^2 + y^2 - 2x - 2y - 23 = 0$
(c) $x^2 + y^2 - 2x + 2y - 23 = 0$
(d) $x^2 + y^2 + 2x - 2y - 23 = 0$

5. The circle $x^2 + y^2 = 9$ is contains in the circle $x^2 + y^2 - 6x - 8y + 25 = c^2$ if **NIMCET 2010**
(a) $c = 2$ (b) $c = 3$
(c) $c = 5$ (d) $c = 10$

6. If $2x + 3y - 6 = 0$ and $9x + 6y - 18 = 0$ cuts the axes in concyclic points, then the centre of the circle is **NIMCET 2011**
(a) (2, 3) (b) (3, 2) (c) (5, 5) (d) (5/2, 5/2)

7. The number of distinct solutions (x, y) of the system of equations $x^2 = y^2$ and $(x - a)^2 + y^2 = 1$, where 'a' is any real number, can only be **NIMCET 2012**
(a) 0, 1, 2, 3, 4 or 5 (b) 0, 1 or 3
(c) 0, 1, 2 or 4 (d) 0, 2, 3 or 4

8. If the circles $x^2 + y^2 + 2x + 2ky + 6 = 0$ and $x^2 + y^2 + 2ky + k = 0$ intersect orthogonally, then k is **NIMCET 2012**
(a) $2 \text{ or } -\frac{3}{2}$ (b) $-2 \text{ or } -\frac{3}{2}$
(c) $2 \text{ or } \frac{3}{2}$ (d) $-2 \text{ or } \frac{3}{2}$

9. Find the equation of the circle which passes through $(-1, 1)$ and (2, 1) and having centre on the line $x + 2y + 3 = 0$ **NIMCET 2013**
(a) $2x^2 + 2y^2 - 2x + 7y - 13 = 0$
(b) $x^2 + 2y^2 - 2x + 7y - 13 = 0$
(c) $2x^2 + 2y^2 + 2x + 7y - 13 = 0$
(d) $x^2 + y^2 + 2x + 7y - 13 = 0$

10. The lines $3x - 4y + 4 = 0$ and $6x - 8y - 7 = 0$ are tangent to the same circle. The radius of this circle is **NIMCET 2013**
(a) 3/2 (b) 3/4 (c) 4/5 (d) 7/10

11. If two circle $x^2 + y^2 + 2gx + 2fy = 0$ and $x^2 + y^2 + 2g'x + 2f'y = 0$ touch each other, then which of the following is true? **NIMCET 2015**
(a) $gf = gf'$ (b) $g'f = gf'$
(c) $gg' = ff'$ (d) None

12. A circle touches the X-axis and also touches another circle with centre at (0, 3) and radius 2. Then the locus of the centre of the first circle is **NIMCET 2015**
(a) a parabola (b) a hyperbola
(c) a circle (d) an ellipse

13. If a twelve side regular polygon is inscribes in a circle of radius 3 cm, then the length of each side of the polygon us **NIMCET 2016**
(a) 3 (b) $18 - 9\sqrt{3}$
(c) $18 + 9\sqrt{3}$ (d) $9(1 - 9\sqrt{3})$

14. The equation of a circle with diameters are $2x - 3y + 12 = 0$ and $x + 4y - 5 = 0$ and area of 154 sq units is **NIMCET 2016**
(a) $x^2 + y^2 - 6x + 4y - 36 = 0$
(b) $x^2 + y^2 + 6x - 4y - 36 = 0$
(c) $x^2 + y^2 - 6x - 4y + 25 = 0$
(d) None of these

15. The point of intersection circle $x^2 + y^2 + 10x - 12y + 51 = 0$ and the line $3y + x = 3$ is **NIMCET 2018**
(a) $(-6, 3)$ (b) $(3, -6)$ (c) $(6, -3)$ (d) $(-3, 6)$

16. The circle whose equations are $x^2 + y^2 + c^2 = 2ax$ and $x^2 + y^2 + c^2 - 2by = 0$ will touch one another externally if **NIMCET 2018**
(a) $\frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a^2}$ (b) $\frac{1}{c^2} + \frac{1}{a^2} = \frac{1}{b^2}$
(c) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$ (d) None of these

17. The equation of the circle passing through the point (4, 6) and whose diameters are along $x + 2y - 5 = 0$ and $3x - y - 1 = 0$ is **NIMCET 2019**
(a) $x^2 + y^2 - 2x - 6y - 20 = 0$
(b) $x^2 + y^2 - 6x - 2y - 20 = 0$
(c) $x^2 + y^2 - 2x - 4y - 20 = 0$
(d) $x^2 + y^2 - 4x - 2y - 20 = 0$

18. For two circles $x^2 + y^2 = 16$ and $x^2 + y^2 - 2y = 0$, there is/are **NIMCET 2019**
(a) one pair of common tangents
(b) two pairs of common tangents
(c) three common tangents
(d) no common tangents

19. Number of common tangnt to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y = 24$ is **NIMCET 2021**
(a) 0 (b) 1 (c) 3 (d) 4

20. A circle touches the x-axis and also touches the circle with centre (0, 3) and radius 2. The locus of the centre of the circle is **NIMCET 2023**
(a) a circle (b) an ellipse
(c) a parabola (d) a hyperbola

21. The circle $x^2 + y^2 + \alpha x + \beta y + \gamma = 0$ is the image of the circle $x^2 + y^2 - 6x - 10y + 30 = 0$ across the line $3x + y = 2$. The value of $\lfloor \alpha + \beta + \gamma \rfloor$ is (where $\lfloor . \rfloor$ represents the floor function) **NIMCET 2025**
(a) 23 (b) 22
(c) 20 (d) 21

22. A circle with its center in the first quadrant touches both the coordinate axes and the line x – y – 2 = 0. Then the area of the circle is **NIMCET 2025**
(a) 2π (b) π (c) $\frac{\pi}{2}$ (d) 4π

ANSWER

1.	a	2.	c	3.	a	4.	c	5.	d
6.	d	7.	d	8.	a	9.	a	10.	b
11.	b	12.	a	13.	b	14.	b	15.	a
16.	c	17.	c	18.	d	19.	b	20.	c
21.	a	22.	a						

Analysis 2008 to 2025
Questions Asked in NIMCET 2008 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	04	NIMCET 2016	02
NIMCET 2008	00	NIMCET 2017	00
NIMCET 2009	00	NIMCET 2018	02
NIMCET 2010	01	NIMCET 2019	02
NIMCET 2011	02	NIMCET 2020	00
NIMCET 2012	01	NIMCET 2021	01
NIMCET 2013	02	NIMCET 2022	00
NIMCET 2014	00	NIMCET 2023	01
NIMCET 2015	02	NIMCET 2024	00
		NIMCET 2025	02

SOLUTION

1. (a) Passing through (2,3) and (-1, 1) with centre lying on the line x – 3y – 11=0
Let circle $x^2 + y^2 + 2gx + 2fy + c = 0$
It passes through (2, 3) (-1, 1) centre (-g, -f)
4 + 9 + 4g + 6f + c = 0
4g + 6f + c = –13 .... (i)
$1^2 + 1^2 + 2g + 2f + c = -2$ .... (ii)
Equation (2) – equation (1)
6g + 4f = –11 ....(iii)
Centre lies in line x – 3g – 11 = 0
Solving (3) and (4^{th}) equation
$g = \frac{-7}{2}, f = \frac{5}{2}$ x = –14
Equation $x^2 + y^2 - 7x + 5y - 14 = 0$

2. (c) Diameter along the line 2x + 3y + 1
3x – y – 4 = 0
Circumference = 10π
circumference = 2πr = 10 π
r = 5
2x + 3y + 1 = 0 ....(1)
3x – y –4 = 0
Equation 1 + equation 2
5x + 2i - 3 = 0 ....(2)
X = 1, y= –1
Centre (1, –1)
Equation of circle
$(x-1)^2 + (y + 1)^2 = (5)^2$
$x^2 + y^2 - 2x + 2y + 2 - 25 = 0$
$x^2 + y^2 - 2x + 2y - 23 = 0$

3. (a) Passing through (2,3) and (-1, 1) with centre lying on the line x – 3y – 11=0
Let circle $x^2 + y^2 + 2gx + 2fy + c = 0$
It passes through (2, 3) (-1, 1) centre (-g, -f)
4 + 9 + 4g + 6f + c = 0
4g + 6f + c = –13 .... (i)
$1^2 + 1^2 + 2g + 2f + c = -2$ .... (ii)
Equation (2) – equation (1)
6g + 4f = –11 ....(iii)
Centre lies in line x – 3g – 11 = 0
Solving (3) and (4^{th}) equation
$g = \frac{-7}{2}, f = \frac{5}{2}$ x = –14
Equation $x^2 + y^2 - 7x + 5y - 14 = 0$

4. (c) Diameter along the line 2x + 3y + 1
3x – y – 4 = 0
Circumference = 10π
circumference = 2πr = 10 π
r = 5
2x + 3y + 1 = 0 ....(1)
3x – y –4 = 0
Equation 1 + equation 2
5x + 2i - 3 = 0 ....(2)
X = 1, y= –1
Centre (1, –1)
Equation of circle
$(x-1)^2 + (y + 1)^2 = (5)^2$
$x^2 + y^2 - 2x + 2y + 2 - 25 = 0$
$x^2 + y^2 - 2x + 2y - 23 = 0$

5. (d) $C_1: x^2 + y^2 - 6x - 8y + 25 = c^2$
$\Rightarrow C_1(3,4), r_1 = c$
$\Rightarrow C_2: x^2 + y^2 = 9$
$\Rightarrow C_2(0,0), r_2 = 3$
For a circle to lie entirely inside the other, then

$C_1C_2 < |r_1 - r_2|$

$C_1C_2 = 5, r_1 - r_2 = c - 3$

$5 < |c - 3|$

From given options, $c = 10$ satisfies the relation.

6. **(d)** $2x + 3y - 6 = 0$

Let this line be $AB, \frac{x}{3} + \frac{y}{2} = 1$

Hence, $A(3,0)$ and $B(0,2)$

And

$9x + 6y - 18 = 0$

Let this line be $CD, \frac{x}{2} + \frac{y}{3} = 1$

Hence, $C(2,0)$ and $D(0,3)$

A, B, C, D are concyclic points.

Let equation of circle be

$x^2 + y^2 + 2gx + 2fy + c = 0$

Hence, these points A, B, C, D satisfy equation of circle.

Solving, we get

$c = 6, f = g = -\frac{5}{2}$

Equation of circle

$x^2 + y^2 - 5x - 5y + 6 = 0$

$\therefore$ Centre $\left(\frac{5}{2}, \frac{5}{2}\right)$

7. **(d)** $x^2 = y^2$ and $(x - a)^2 + y^2 = 1$

$C(a, 0) \Rightarrow R = 1$

$y^2 = x^2$

$y = \pm x$

$(x - a)^2 + x^2 = 1$

$2x^2 + a^2 - 2ax = 1$

$2x^2 - 2ax + (a^2 - 1) = 0$

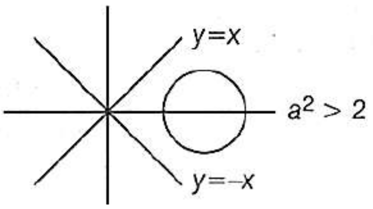
$\therefore\ x = \frac{2a \pm \sqrt{4a^2 - 8(a^2 - 1)}}{4}$

$= \frac{2a \pm \sqrt{8 - 4a^2}}{4}$

$= \frac{a \pm \sqrt{2 - a^2}}{2}$

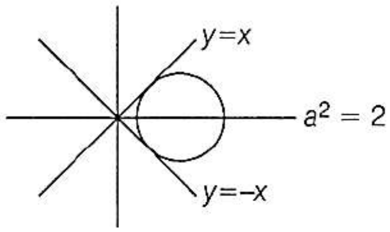
So, the number of points of intersection will depend upon the value of a.

Case 1: Circle will not intersect the lines.



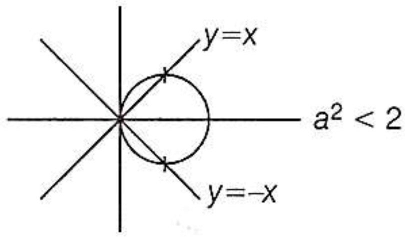
No solution.

Case 2: If circle touches the lines.



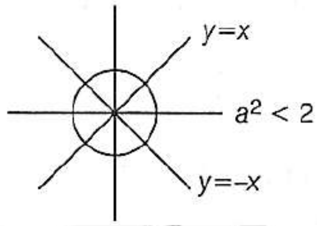
Two solutions.

Case 3: If circle passes through origin.



Three solutions.

Case 4: For four solutions



$\therefore$ These can be 0,2,3,4 distinct solutions.

8. **(a)**

$x^2 + y^2 + 2x + 2ky + 6 = 0, x^2 + y^2 + 2ky + k = 0$

$C_1(-1, -k)$

$r_1 = \sqrt{1 + k^2 - 6} = \sqrt{k^2 - 5}$

$C_2(0, -k)$

$r_2 = \sqrt{k^2 - k}$

For orthogonal intersection

$r_1^2 + r_2^2 = (C_1C_2)^2$

$(k^2 - 5) + (k^2 - k) = 1$

$\Rightarrow\ 2k^2 - k - 6 = 0$

$\Rightarrow\ 2k^2 - 4k + 3k - 6 = 0$

$\Rightarrow\ 2k(k - 2) + 3(k - 2) = 0$

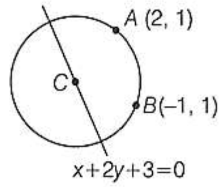
$\Rightarrow\ k = \frac{-3}{2}, k = 2$

9. **(a)** Let (α, β) is the centre of circle which lies on the line

$x + 2y + 3 = 0$

$\Rightarrow\ \alpha + 2\beta + 3 = 0 \Rightarrow \beta = \frac{-\alpha - 3}{2}$

Hence, centre $C\left(\alpha, \frac{-\alpha - 3}{2}\right)$



$CA = CB =$ radii

$CA^2 = CB^2$

$\Rightarrow (\alpha-2)^2 + \left(\frac{-\alpha-3}{2} - 1\right)^2 = (\alpha+1)^2 + \left(\frac{-\alpha-3}{2} - 1\right)^2$

$\Rightarrow \alpha^2 + 4 - 4\alpha = \alpha^2 + 1 + 2\alpha$

$\Rightarrow 6\alpha = 3$

$\Rightarrow \alpha = \frac{1}{2}$

$\therefore C\left(\frac{1}{2}, \frac{-7}{4}\right)$

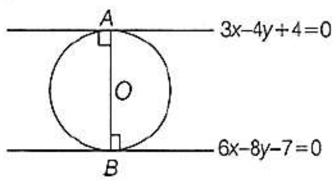
Radius $= \sqrt{\left(2-\frac{1}{2}\right)^2 + \left(1+\frac{7}{4}\right)^2} = \frac{\sqrt{157}}{4}$

Equation $\left(x-\frac{1}{2}\right)^2 + \left(y+\frac{7}{4}\right)^2 = \frac{157}{16}$

$x^2 + \frac{1}{4} - x + y^2 + \frac{49}{16} + \frac{7}{2}y = \frac{157}{16}$

$2x^2 + 2y^2 - 2x + 7y - 13 = 0$

10. **(b)** AB is diameter



$L_1 = 3x - 4y + 4 = 0$

$L_2 = 3x - 4y - 7/2 = 0$

$AB = \left|\frac{4-\left(\frac{-7}{2}\right)}{\sqrt{9+16}}\right| = \left|\frac{15}{2\times5}\right| = \frac{3}{2}$ $\left[\because d = \frac{C_1 - C_2}{\sqrt{a^2+b^2}}\right]$

Hence, radius is 3/4.

11. **(b)** $x^2 + y^2 + 2gx + 2fy = 0$

$C_1(-g,-f), r_1 = \sqrt{g^2+f^2}$

$x^2 + y^2 + 2g'x + 2f'y = 0$

$C_2(-g',-f'), r_2 = \sqrt{g'^2+f'^2}$

When two circle touches, $r_1 + r_2 = C_1C_2$

$\sqrt{g^2+f^2} + \sqrt{g'^2+f'^2} = \sqrt{(g-g')^2 + (f-f')^2}$

Squaring both sides, we get

$g^2 + g'^2 + f^2 + f'^2 + 2\sqrt{g^2+f^2}\sqrt{g'^2+f'^2}$

$= g^2 + g'^2 - 2gg' + f^2 + f'^2 - 2ff'$

$(g^2+f^2)(g'^2+f'^2) = g^2g'^2 + f^2f'^2 + 2gg'ff'^2$

$g^2f'^2 + f^2g'^2 = 2gg'ff'$

$(gf' - fg')^2 = 0$

$gf' = fg'$

12. **(a)** Circle with centre $C_1(h,k)$ touches X-axis.

$\therefore$ Radius $= k = r_1$

Now, since this circle touches circle with $C_2(0,3)$ and $r_2 = 2$

$\therefore C_1C_2 = r_1 + r_2$

$h^2 + (k-3)^2$

$= (2+k)^2$

$h^2 + k^2 + 9 - 6k$

$= (2+k)^2$

$h^2 + k^2 + 9 - 6k$

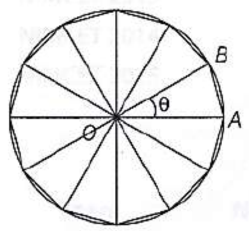
$= 4 + k^2 + 4k$

$h^2 = 10k - 5$

$x^2 = 10y - 5$

which is a parabola.

13. **(b)** Since, it is a regular polygon (it has 12 sides).



Each side will subtend an equal $\angle\theta$ at centre

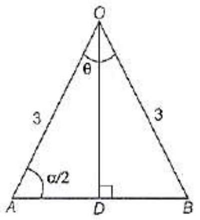
$12\theta = 360°$

$\Rightarrow \theta = 30°$

$OA = OB =$ Radius $= 3$ cm

Sum of internal angles of a polygon $= (n-2)180°$

Let the internal angle be α.



Hence, $(12-2)\times 180° = 12\alpha$

$\Rightarrow \alpha = 150°$

$\therefore \angle A = \angle B = \frac{\alpha}{2} = 75°$

$\therefore \cos 75° = \frac{AD}{3}$

$\Rightarrow AD = 3\cos 75°$

$\therefore AB = 6\cos 75°$

$= 6 \times \frac{(\sqrt{3}-1)}{2\sqrt{2}} = \frac{3}{\sqrt{2}}(\sqrt{3}-1) > 0$

Since, $AB < 3$

$\frac{3}{\sqrt{2}}(\sqrt{3}-1) < 3$

Here, exact answer does not match.

From given options only (b) $18 - 9\sqrt{3}$ satisfies the result.

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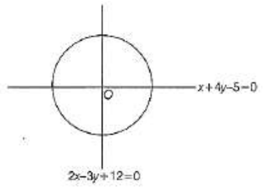
14. (b) Centre is the intersection point of the diameters.

$\therefore\ 2x-3y+12=0$

$2x+8y-10=0$



Solving these two equations, we get

Centre $O(-3,2)$

Given, $A=\pi r^2=154$

$\Rightarrow\ \frac{22}{7}\times r^2=154\Rightarrow r=7$ units.

$\therefore$ Equation, $(x+3)^2+(y-2)^2=49$

$x^2+y^2+6x-4y-36=0$

15. (a) $S: x^2+y^2+10x-12y+51=0,$

$L: 3y+x=3$

$x=3-3y$

$\Rightarrow\ 9+9y^2-18y+y^2+30-30y-12y+51=0$

$\Rightarrow\ 10y^2-60y+90=0$

$\Rightarrow\ y^2-6y+9=0$

$\Rightarrow\ (y-3)^2=0$

$\Rightarrow\ y=3$

$\Rightarrow\ x=3-3(3)$

$\Rightarrow\ x=-6$

$\therefore P(-6,3)$

16. (c)

$S_1: x^2+y^2-2ax+c^2=0, x^2+y^2-2by+c^2=0$

$C_1(a,0)$ $\qquad$ $C_2(0,b)$

$r_1=\sqrt{a^2-c^2}$ $\qquad$ $r_2=\sqrt{b^2-c^2}$

$r_1=\sqrt{a^2-c^2}\ r_2=\sqrt{b^2-c^2}$

S_1 and S_2 touches externally

$\therefore C_1C_2=r_1+r_2$

$\sqrt{a^2+b^2}=\sqrt{a^2-c^2}+\sqrt{b^2-c^2}$

$\Rightarrow a^2+b^2=a^2-c^2+b^2-c^2+2\sqrt{a^2-c^2}\sqrt{b^2-c^2}$

$\Rightarrow 2c^2=2\sqrt{a^2-c^2}\sqrt{b^2-c^2}$

$\Rightarrow c^4=(a^2-c^2)(b^2-c^2)$

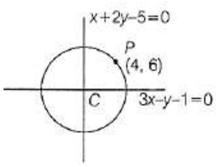
$\Rightarrow\ c^4=a^2b^2-a^2c^2-b^2c^2+c^4$

$\Rightarrow\ a^2b^2=a^2c^2+b^2c^2$

$\Rightarrow \frac{1}{c^2}=\frac{1}{a^2}+\frac{1}{b^2}$

17. (c) Centre 3x + 6y – 15 = 0, 3x – y – 1 = 0

y = 2 and x = 1



Centre (1,2)

Radius = CP = $\sqrt{(4-1)^2+(6-2)^2}=\sqrt{9+16}=5$

Equation $(x-1)^2+(y-2)^2=25$

$\Rightarrow\ x^2+y^2-2x-4y-20=0$

18. (d) Circle $S_1, x^2+y^2=16$

Circle $S_2, x^2+y^2-2y=0$

Centre C_1 is $C_1(0,0)$ and radius $r_1=4$

Centre C_2 is $C_2(0,1)$ and radius $r_2=1$

Distance between centres is $C_1C_2=1$

Since, $C_1C_2<|r_2-r_1|$

Circle S_2 is completely inside S_1

$\therefore$ No common tangents

19. (b) $S_1: x^2+y^2=4, S_2: x^2+y^2-6x-8y=24$

$C_1(0,0)\ C_2(3,4)$

$r_1=2\ r_2=7$

$\therefore C_1C_2=|r_1-r_2|=5$

$\because$ Circle touches internally.

$\therefore$ Number of common tangents = 1

20. (c) Refer to solution 8.

21. (a)

Original circle: $x^2+y^2-6x-10y+30=0$.

Complete the square: $(x-3)^2+(y-5)^2=4$.

So centre $C=(3,5)$ and radius $r=2$.

Line: $3x+y=2$ or $3x+y-2=0$.

Reflect point (x_0,y_0) across $ax+by+c=0$ by

$$(x',y')=\left(x_0-\frac{2a(ax_0+by_0+c)}{a^2+b^2},\ y_0-\frac{2b(ax_0+by_0+c)}{a^2+b^2}\right)$$

Here $a=3,\ b=1,\ c=-2$. For $C=(3,5)$:

$$ax_0+by_0+c=3\cdot3+1\cdot5-2=12,\quad a^2+b^2=10,$$

so

$$C'=\left(3-\tfrac{2\cdot3\cdot12}{10},\ 5-\tfrac{2\cdot1\cdot12}{10}\right)=\left(-\tfrac{21}{5},\ \tfrac{13}{5}\right).$$

Image circle has same radius $r=2$. Equation:

$$\left(x+\tfrac{21}{5}\right)^2+\left(y-\tfrac{13}{5}\right)^2=4.$$

Expand to $x^2+y^2+\alpha x+\beta y+\gamma=0$. Calculating,

$$\alpha=\tfrac{42}{5},\qquad \beta=-\tfrac{26}{5},\qquad \gamma=\left(\tfrac{21}{5}\right)^2+\left(\tfrac{13}{5}\right)^2-4=\tfrac{102}{5}.$$

So

$$\alpha+\beta+\gamma=\tfrac{42-26+102}{5}=\tfrac{118}{5}=23.6.$$

Therefore $\lfloor\alpha+\beta+\gamma\rfloor=23$.

Answer: (a) 23.

22. (a)

- Let center be $(a,a)\rightarrow$ radius = a (since it touches both axes).

- Distance from (a,a) to line $x-y-2=0$ must be a:

$$\frac{|a-a-2|}{\sqrt{1^2+(-1)^2}} = a \rightarrow \frac{2}{\sqrt{2}} = a \rightarrow a = \sqrt{2}$$

- Area $= \pi a^2 = \pi\left(\sqrt{2}\right)^2 = 2\pi$

Ans. 2π